WiseChoice: An Intelligent Workspace for Online Shopping Decision-Making

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***Abstract*—Current e-commerce interfaces exacerbate the "paradox of choice" by presenting unstructured, redundant, and jargon-heavy product information, leading to significant cognitive overload. While Large Language Models (LLMs) offer conversational capabilities, they often fail to provide the structured, comparative reasoning required for online shopping decision-making. We introduce *WiseChoice*, an intelligent workspace acting as an "Online Shopping Copilot" that transforms raw product pages into structured decision reports. To evaluate its efficacy, we conducted a within-subjects study ($N = 12$ product pairs) assessing three causal dimensions of user experience: reading load, decision efficiency, and terminology comprehension. Results indicate that WiseChoice significantly reduces cognitive load, lowering the estimated reading time by 56% ($t(11) = 9.35, p < .001$) and compressing decision-relevant information by 84.5% (RR_{diff}). Furthermore, the system achieved a Terminology Barrier Rate of 0%, effectively democratizing expert-level decision-making for novice users. This work contributes empirical evidence on how constrained generative agents can reshape information consumption from passive scrolling to active synthesis.**

***Index Terms*—Generative AI, Decision Support, Cognitive Load, E-commerce, Human-AI Collaboration**

I. INTRODUCTION

The digitalization of commerce has drastically reduced the cost of information access but significantly increased the cost of information processing. A typical product detail page (PDP) on platforms like Amazon contains thousands of words, amalgamating critical technical specifications with marketing rhetoric and redundant descriptions. hen the intrinsic load of processing this unstructured information exceeds a user's working memory capacity, "decision paralysis" occurs. Users are forced to manually scan, memorize, and compare high-dimensional attributes (averaging 64 attributes per category [1]) across disparate browser tabs, a process that is both error-prone and mentally exhausting.

Current solutions fall short of addressing this structural problem. Faceted search filters are rigid and fail to capture semantic nuances, while general-purpose conversational agents (e.g., ChatGPT) often provide hedged, unstructured advice (e.g., "Both products are good") that fails to reduce the decision space. There is a lack of systems that can act not just as information retrievers, but as *decision analysts* that synthesize raw data into actionable insights.

To bridge this gap, we present **WiseChoice**, a Human-AI collaborative workspace designed to act as a "Shopping Copilot." The system shifts the user experience from "scrolling through multiple long pages" to "reading a single structured decision report." WiseChoice ingests heterogeneous e-commerce data and utilizes a constrained LLM agent to generate reports focusing on key differentiators and terminology explanation.

In this work, we articulate our system design and present a quantitative evaluation addressing the following Research Question (RQ):

> *How does a structured, LLM-driven decision report affect reading load, decision efficiency, and terminology comprehension compared to standard e-commerce interfaces?*

Our contributions are twofold: (1) The design of *WiseChoice*, featuring a novel "difference-only" prompting strategy; and (2) Empirical evidence demonstrating significant reductions in cognitive load and terminology barriers.

II. TECHNICAL APPROACH

A. System Overview

WiseChoice is architected as a modular intelligent workspace that integrates directly into the user's browsing workflow. The system comprises three core components: an interactive *Dual-Mode Workspace* for frontend interaction, a *Data Connector* for information ingestion and normalization, and a backend *Decision Assistant Agent* for cognitive synthesis. Figure 1 illustrates the user interface, highlighting the transition from candidate management (List Mode) to structured decision support (Report Mode).

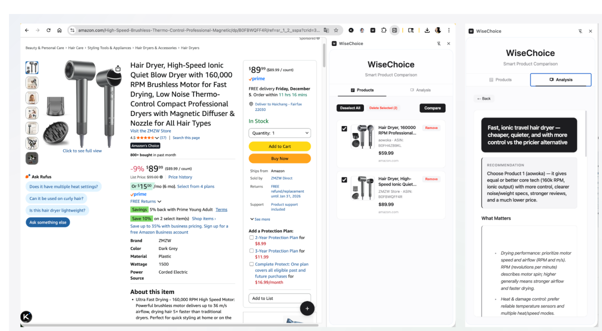

Fig. 1. The WiseChoice Dual-Mode Workspace. Left: **List Mode** allows users to curate candidate products with minimal distraction. Right: **Report Mode** presents the AI-generated structured decision analysis, featuring the Recommendation, Strategy, and Key Differences cards.

B. Scenario

When making online purchases of high-value or complex products, users often need to open multiple browser tabs, manually record key parameters, and repeatedly switch between pages for comparison. This process is fragmented, inefficient, and can easily lead to cognitive overload. Furthermore, users may not understand the technical terminology associated with certain products, which increases the cost of decision-making and negatively impacts the user experience. WiseChoice aims to fundamentally simplify this process by providing an integrated, AI-driven decision support tool, freeing users' time and mental energy from tedious information comparison and searches for unfamiliar information, allowing them to focus on higher-level decision trade-offs.

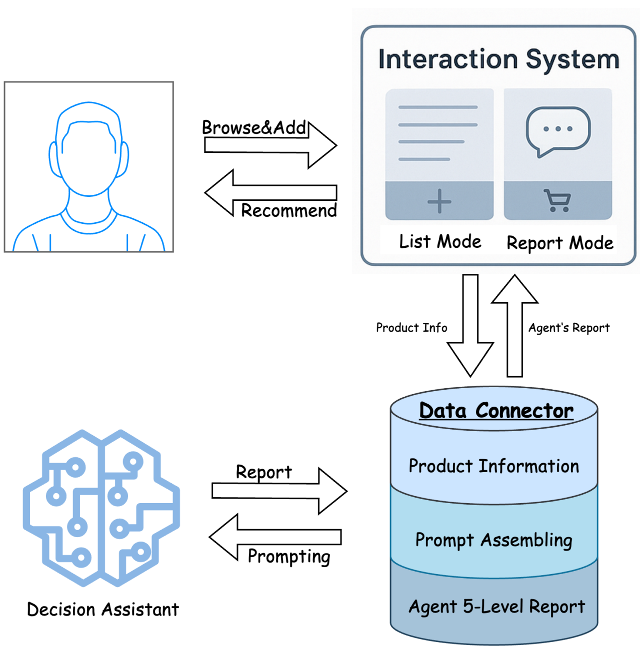

Fig. 2. System workflow of WiseChoice. The diagram illustrates the interaction between the user and three core components: (1) Interaction System — composed of List Mode and Report Mode interfaces; (2) Data Connector — bridging the interface and backend; and (3) Decision Assistant — the local AI agent generating the five-level report.

C. Component #1: Interactive Dual-Mode Workspace

As a user browses potential television options, they first interact with the Dual-Mode Workplace. WiseChoice is a Chrome browser extension designed to reduce the decision-making costs for users facing information overload or unfamiliarity with product details during online shopping. Its core is a Split-Pane AI Decision Workplace, which reframes the browser viewport into two independent workspaces: a content browsing area and an AI decision-making area. Users can immersively browse products on the left while efficiently managing, comparing, and making decisions about selected items using AI decision assistant in the dedicated area on the right. The Workplace features two seamlessly switchable interaction modes: List Mode, which serves as the interface for information aggregation where users can add products to a list via a floating "+" button injected by a content script; and Report Mode, the interface for analysis and decision-making. When a user selects products and clicks "compare," this mode simulates a report interface to present the AI's analysis and recommendations.

a) Implementation Plan: This component is built using native JavaScript (ES6+), native DOM APIs, and CSS Flexbox, deeply integrated with the Chrome Extension API (Manifest V3) to ensure minimal performance overhead. The workspace is implemented with static assets (`sidebar.html`, `sidebar.css`, `sidebar.js`), and its visibility is managed via `chrome.sidePanel.open` as declared in the manifest. The core logic resides within `sidebar.js`, which directly manages application state—such as the `candidates` array and `selectedIds` set—and handles view switching between List and Report modes by toggling CSS classes on DOM containers.

For data collection, a content script (`content.js`) containing specific DOM selector rules extracts product metadata from e-commerce sites. All internal communication uses the `chrome.runtime.sendMessage` API to send messages to the background script, `background.js`. Crucially, in line with Manifest V3, this background script functions as a non-persistent Service Worker, acting as an event-driven handler that awakens only when needed to manage network requests to the local backend API.

D. Component #2: Data Connector

The Data Connector serves as the data orchestration layer of the application, positioned as a stateless intermediary layer that bridges the user-facing Dual-Mode Workspace with the backend Decision Assistant. This component is not directly visible to the user but is critical for orchestrating the flow of data and logic. Its primary responsibility is to translate abstract user requests from the frontend into precisely structured, actionable instructions for the AI. This process involves three key functions: first, it ingests and sanitizes the raw product data sent from the browser extension; second, it performs the crucial task of prompt engineering, assembling the clean data into a comprehensive prompt tailored for the AI agent; and third, it receives the agent's structured JSON output and routes it back to the workspace. This architectural decoupling is paramount, as it allows the frontend presentation logic and the backend cognitive logic to evolve independently, enhancing modularity, testability, and overall system maintainability.

a) Implementation Plan: The Data Connector is implemented as a local, asynchronous server using Python 3, built upon the high-performance `FastAPI` web framework to handle concurrent requests efficiently. It exposes a specific API endpoint, `/api/v1/agent/decision`, to which the browser extension sends POST requests. Upon receiving a request, data integrity and structure are rigorously enforced by `Pydantic` models that validate the incoming product information against a predefined schema. Following successful validation, a dedicated prompt engineering module formats the data and injects it into a system prompt template. This template contains the role-playing instructions, the context of the user's task, and an explicit schema for the desired JSON output. After the Decision-making Agent completes its analysis, the FastAPI application packages the resulting object into an HTTP response. This response is the final payload, structured to be directly transformed by the Dual-Mode Workspace into the five-level decision report for the user.

E. Component #3: Decision Assistant Agent

When the user selects multiple products from the List Mode in Workspace and initiates a comparison, they will enter Report Mode and interact with the Decision Assistant. This component is the intelligent core of the system, driving the `GPT-5-mini` API with prompt engineering to perform comparative analysis. Its role is constrained to execute the inference task based solely on the complete prompt provided by the Data Connector. It transforms disparate product specifications into a single, coherent, and human-readable analysis. The agent's primary objective is to produce not freeform text, but a structured, multi-faceted decision report. This output is precisely formatted to align with the five-level report structure displayed in the frontend's Report Mode, encompassing: a definitive Documentary "Title", a concise "Recommendation summary", an explanation of "What Matters" for the product category, a comparative analysis of "Key Differences", and the final "Why This Choice" justification.

a) Implementation Plan: This component is implemented as a modular Python function invoked by the Data Connector. It receives the fully assembled prompt as its sole input, abstracting it from the complexities of data collection and user interaction. The agent uses a standard HTTP client library, such as `httpx`, to make a synchronous API call to the `GPT-5-mini` model endpoint, passing the prompt as the payload. The prompt explicitly instructs `GPT-5-mini` to return its response as a JSON-formatted string. The agent's final task is to parse this string to validate its structure before returning the resulting Python dictionary to the Data Connector. This design keeps the agent lightweight and focused exclusively on the core task of interacting with the language model.

III. Evaluation Design

To rigorously assess the causal impact of WiseChoice on the online shopping experience, we conducted a controlled, system-level quantitative evaluation. The study was designed to prioritize generalization across product categories, aiming to establish causal links between the intervention (structured AI reports) and improvements in reading load, decision efficiency, and terminology comprehension.

A. Research Goal and Hypotheses

The primary objective of this study was to investigate whether replacing standard e-commerce interfaces with AI-synthesized reports reduces cognitive friction. We formulated three specific hypotheses to guide our investigation. First, we hypothesized that WiseChoice would significantly reduce text volume and information redundancy compared to raw product pages (H1: Reading Load). This hypothesis posits that unstructured marketing copy constitutes a significant portion of the cognitive burden. Second, we posited that by filtering out similarities and focusing exclusively on key differences, WiseChoice would improve decision efficiency (H2: Decision Efficiency). Finally, we hypothesized that WiseChoice would effectively minimize terminology barriers by providing in-context explanations for technical jargon (H3: Terminology Barriers), thereby bridging the gap between novice and expert users.

B. Experimental Setup

We employed a within-subjects design where each product pair served as its own control. This methodology was chosen to strictly eliminate confounds arising from category-specific complexity, page length variations, or brand-specific marketing styles. The dataset consisted of $N = 12$ pairs of competing products (24 items total) randomly sampled from diverse high-volume Amazon categories, including Electronics (e.g., Laptops, Cameras, Monitors), Home & Kitchen (e.g., Appliances, Vacuums), and Personal Care.

For each pair, we established two distinct conditions for evaluation. The Control condition consisted of the raw product page text, encompassing titles, bullet points, product descriptions, and technical specification tables. This represents the baseline information environment a user navigates today. The Treatment condition consisted of the full WiseChoice decision report generated for the same product pair. No human participants were involved in this phase; instead, the experiment utilized a system-level analysis of outputs under controlled conditions to validate information-theoretical improvements and structural efficiency.

C. Measures and Operationalization

To quantify the user experience improvements, we operationalized our hypotheses using five objective, ratio-scale metrics.

To assess **Reading Load (H1)**, we calculated the *Estimated Reading Time* as the total word count divided by 238 words per minute (wpm). This value is derived from the standardized average English silent reading rate established by Brysbaert [2], providing a consistent baseline for temporal cost. We also computed the *Full Report Information Reduction Ratio (RR_{full})*, defined as the percentage reduction in total reading volume between the raw product page and the WiseChoice full report. This metric serves as a proxy for the reduction in overall cognitive entry barriers.

For **Decision Efficiency (H2)**, we focused on the information density of the decision-making process. We introduced the *Key Differences Information Reduction Ratio (RR_{diff})*, which measures the percentage reduction in reading volume when users focus strictly on the "Key Differences" section of the report compared to the raw text. This metric quantifies the efficiency gain for users specifically seeking to distinguish

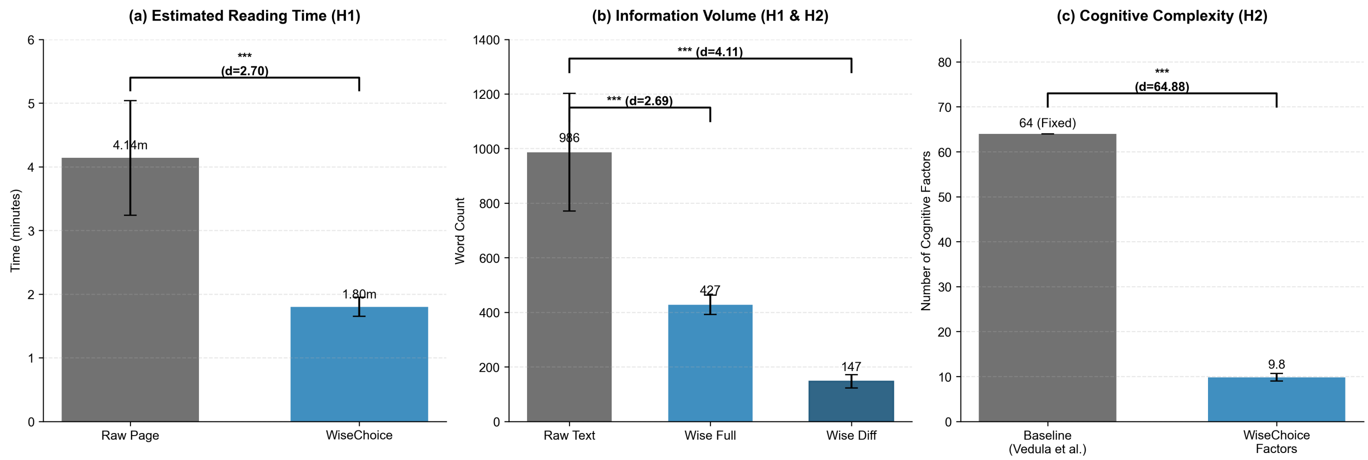

Fig. 3. Evaluation Results ($N = 12$). (a) Comparison of Estimated Reading Time showing significant reduction and lower variance in the WiseChoice condition. (b) Decision Efficiency metrics showing high compression of textual differences (RR_{diff}) and cognitive factors ($CFCR$).

between options. Additionally, we calculated the *Cognitive Factor Compression Rate (CFCR)*, defined as the ratio of key attributes presented in the WiseChoice report to a fixed baseline of 64 attributes. This baseline represents the average number of comparable product attributes across Amazon categories, as identified by Vedula et al. [1]. A lower CFCR indicates a more manageable decision space.

Finally, to evaluate **Terminology Barriers (H3)**, we measured the *Terminology Barrier Rate (TBR)*. This is defined as the proportion of identified technical terms in the WiseChoice output that remain unexplained. The target value for this metric is 0, based on the assumption that proprietary or technical terms on raw pages are inherently unexplained for laypersons and thus constitute a barrier to understanding.

D. Analysis Plan

To establish causality, we conducted paired statistical tests comparing the Control and Treatment conditions for each measure. A paired samples t-test was used as the primary analysis for Estimated Reading Time, RR_{full}, and RR_{diff} to determine statistical significance, assuming the distribution of differences met normality assumptions. For the Cognitive Factor Compression Rate (CFCR), since we are comparing the distribution of WiseChoice factors against a fixed literature baseline (constant = 64), we employed the non-parametric Wilcoxon signed-rank test. This robust test evaluates whether the median number of cognitive factors presented by WiseChoice was significantly lower than the industry average. For TBR, we report descriptive statistics compared against the baseline assumption of 100% barriers in the control condition. Effect sizes were calculated using Cohen's d for parametric tests to quantify the magnitude of the intervention's impact.

IV. Results

Our quantitative analysis of the 12 product pairs ($N = 12$) reveals significant and consistent improvements across all measured dimensions. The results in Figure 3 strongly support all three hypotheses, demonstrating that the structured intervention fundamentally alters the information landscape for the user.

A. H1: Reduction in Reading Load

A paired-samples t-test revealed a statistically significant difference in the *Estimated Reading Time* between conditions, $t(11) = 9.35, p < .001$. Participants using WiseChoice would require significantly less time ($M = 1.80$ min$, SD = 0.15$) to process the decision-relevant information compared to the Raw condition ($M = 4.14$ min$, SD = 0.90$).

The magnitude of this difference is substantial, represented by a large effect size of Cohen's $d = 2.70$. This indicates that the reduction is not merely statistically significant but practically transformative—users are saving more than half of the time typically spent on information gathering.

More importantly, the analysis of variance reveals a critical insight: the standard deviation in the Control condition ($SD = 0.90$) was six times higher than in the WiseChoice condition ($SD = 0.15$). This suggests that while raw product pages vary wildly in length and complexity, WiseChoice successfully normalizes the information consumption experience. Regardless of the input's verbosity, the system provides a predictable, low-variance temporal cost for decision-making.

In terms of total information volume, the system achieved a mean *Full Report Information Reduction Ratio* (RR_{full}) of 54.58% ($SD = 10.9\%$). This represents a substantial compression of the marketing content, with a large effect size of $d = 2.69$, indicating that the system effectively condenses more than half of the text without sacrificing decision utility.

B. H2: Enhancement of Decision Efficiency

The system demonstrated a profound ability to filter noise. The *Key Differences Information Reduction Ratio* (RR_{diff}) reached a mean of 84.52% ($SD = 3.85\%$), with an extremely large effect size of $d = 4.11$. This finding implies that when a user's goal is specifically to differentiate between two products, over 84% of the content on a standard e-commerce page is effectively redundant. WiseChoice acts as a high-precision semantic filter, distilling the information down to the critical 15% signal required for comparison.

Regarding cognitive complexity, the system extracted an average of 9.83 cognitive factors per product pair ($SD = 0.83$). A Wilcoxon signed-rank test confirmed that this was significantly lower than the industry baseline of 64 attributes ($Z = -3.06, p < .001$). The resulting effect size was massive ($d = 64.88$). In practical terms, this signifies a 6.5-fold reduction in the dimensionality of the decision space. Users are no longer required to hold dozens of disparate attribute values in working memory; instead, they can focus on a manageable set of approximately 10 synthesized decision points (e.g., "Battery Life" vs. "Portability"), shifting the cognitive task from memorization to judgment.

C. H3: Elimination of Terminology Barriers

An expert audit of the generated reports identified 48 unique technical terms across the dataset (e.g., "Nits", "Micellar", "kPa", "TDP"). Ideally, a user-centric system should bridge the gap between technical specifications and user understanding. Our results confirm this capability: the system successfully provided in-context explanations for all 48 identified terms, resulting in a mean **Terminology Barrier Rate (TBR) of 0.00%**. Unlike the baseline condition where such terms often appear without context, WiseChoice consistently embedded definitions directly into the decision logic, effectively serving as a just-in-time domain educator.

V. Discussion

A. From Information Retrieval to Cognitive Offloading

The exceptionally high RR_{diff} (84.5%) and large effect size ($d = 4.11$) suggest a fundamental shift in the cognitive demands of online shopping. Traditional e-commerce interfaces force users to perform "information retrieval" and "integration" simultaneously—scanning dense text while mentally maintaining a comparison matrix. WiseChoice effectively performs *cognitive offloading* by externalizing these high-cost cognitive processes to the generative agent. By programmatically enforcing a "difference-only" constraint, the system restructures the information architecture from a comprehensive listing (which maximizes recall) to a comparative synthesis (which maximizes precision). This aligns with the principle of supporting *bounded rationality*, allowing users to allocate their limited cognitive resources to evaluating trade-offs rather than filtering noise.

B. Democratizing Domain Expertise via In-Context Learning

The perfect TBR score (0%) indicates that LLMs can serve as effective *just-in-time educators*. In traditional interfaces, encountering an unfamiliar term (e.g., "OLED" or "TDP") forces a context switch—the user must leave the decision flow to search for a definition. WiseChoice embeds these definitions directly into the decision logic (e.g., "Choose 120Hz *for smoother motion*"). This design pattern transforms the decision-making process into a micro-learning opportunity, progressively upgrading the user's domain knowledge without breaking their flow. This suggests a future where e-commerce interfaces do not merely sell products but actively elevate user expertise.

C. Limitations and Future Work

While our quantitative metrics demonstrate significant efficiency gains, the definition of "jargon" remains partially subjective. A term considered basic by an expert might still be a barrier for a complete novice. Future iterations could explore *adaptive complexity*, where the system tailors the granularity of explanations based on a user's inferred knowledge level. Additionally, while objective efficiency is crucial, future work should include a user study with human participants to measure subjective satisfaction, trust in the AI's recommendations, and decision confidence.

VI. Conclusion

WiseChoice demonstrates that Generative AI, when constrained by rigorous design principles, can effectively deconstruct the complexity of online shopping. By reducing reading loads by over 50% and decision noise by 85%, the system provides a blueprint for the next generation of "AIUI" commerce interfaces—where the focus shifts from managing information to making wise choices.

References

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